

Exact Bayesian Inference for Spike-and-Slab Priors in Takagi–Sugeno–Kang Fuzzy Systems with Approximately Calibrated Model-Averaged Prediction Intervals

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Abstract

Calibrated prediction intervals are essential for decision-making under uncertainty, and Bayesian model selection is a principled route to them for rule-based regressions with linear consequents. However, replacing posterior sampling with an analytical approximation—BIC-based inclusion probabilities, hard thresholding, and a Laplace predictive covariance—can collapse accuracy and, in the implementation we study, coverage on ill-conditioned design matrices. We introduce exact inference (up to finite Monte Carlo error) for spike-and-slab priors over Takagi–Sugeno–Kang (TSK) consequents: a block-Gibbs sampler whose per-rule inclusion full conditional is computed in $O(n(d+1)^2)$ time via matrix-determinant and Woodbury identities, combined with Bayesian model averaging (BMA) that propagates model uncertainty into the predictive variance. On two UCI benchmarks (three targets), this yields near-nominal 95% intervals (PICP = 0.93–0.95) where the analytical approximation we study undercovers (PICP = 0.00–0.18), at interval widths comparable to the closed-form conjugate baseline’s on the Energy targets; a corrected dense baseline—a membership-function fix raising R^2 from 0.48 to 0.94 on the Energy-Cooling target—anchors the comparison. The approach applies to any rule-based regression with linear consequents, and the code and corrected baselines are released for reproducible, uncertainty-aware inference.

Keywords: uncertainty quantification, prediction intervals, Bayesian inference, spike-and-slab prior, Bayesian model averaging, Takagi–Sugeno–Kang fuzzy system

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1. Introduction

Predictive models in engineering increasingly need more than point estimates: designers require calibrated uncertainty to support decisions under limited data. Rule-based regressions with linear consequents—of which Takagi–Sugeno–Kang (TSK) fuzzy systems [1] are a canonical instance—are attractive here because they combine nonlinear approximation with a compact, interpretable architecture, and because their linear consequents make Bayesian treatment especially natural, yielding posterior distributions, prediction intervals, and principled model selection. Building-energy simulation exemplifies the data regime that motivates this work: each simulation run is computationally expensive, yielding training sets of only hundreds of samples [2, 3].

The Bayesian TSK literature has been hampered by two problems. The first is methodological. Spike-and-slab priors [4, 5] place positive probability on exact zero coefficients, and a natural alternative to Markov chain Monte Carlo (MCMC) is analytical model selection—BIC-based inclusion probabilities followed by hard thresholding and a Laplace approximation to the predictive covariance. This approximation fails in two ways. Hard thresholding at posterior inclusion probability 0.5 discards rules that carry real (if partial) signal and collapses accuracy; and the pipeline as commonly implemented—a ridge-regularized fit of the ill-conditioned design matrix combined with a Laplace predictive covariance on the pruned model—produces prediction intervals that cover only 0–18% of test points at a nominal 95% level. A faithful version of thresholding plus the Laplace covariance, fit by ordinary least squares, covers near-nominally but remains inaccurate and inefficient (Supplementary, Section 5). The approximation thus sacrifices accuracy, calibration, or both in exchange for a sparsity that, on the relevant benchmark, is not warranted.

The second problem is a *reproducibility* one: a subtle implementation error in the construction of fuzzy membership functions propagates through every downstream baseline. Specifically, an implementation pattern that computes the spreads of Gaussian membership functions from the training data during model fitting, but then *recomputes* them from query data (with incorrect

cluster labels) at prediction time, silently depresses dense TSK baselines, because the fuzzy partition at prediction time no longer matches the one learned during training. We reproduce this pattern, quantify its effect, and show that correcting the bug raises the dense TSK least-squares baseline on the Energy Efficiency cooling target from $R^2 = 0.48$ to 0.94 (Section 5.1), providing a concrete corrected benchmark against which the methodological contribution below is validated.

We replace the analytical approximation with exact MCMC inference. We implement a block-Gibbs sampler for the spike-and-slab prior and predict by Bayesian model averaging (BMA), in which the predictive distribution is a mixture over posterior draws rather than a point estimate from a single selected model. Averaging over rule configurations retains partial information from uncertain rules and carries model uncertainty into the predictive variance, which avoids both the accuracy collapse and the interval collapse. We show that this recovers near-nominal prediction intervals, $\text{PICP} = 0.93\text{--}0.95$ at the nominal 95% level, where the analytical approximation it replaces undercovers severely as implemented ($\text{PICP} = 0.00\text{--}0.18$), at interval widths comparable to the closed-form conjugate baseline’s on the Energy targets. To our knowledge, no prior work provides exact posterior sampling for spike-and-slab priors over rule-model consequents together with model-averaged prediction intervals.

Our contributions are threefold:

1. **Exact spike-and-slab inference for linear-consequent rule models.** A block-Gibbs sampler whose per-rule inclusion full conditional is computed in $O(n(d+1)^2)$ time, with Bayesian model averaging (BMA) propagating model uncertainty into the predictive variance.
2. **Calibrated model-averaged prediction intervals and a diagnostic of analytical approximations.** Gibbs sampling with BMA recovers near-nominal 95% intervals ($\text{PICP} = 0.93\text{--}0.95$), in contrast to our implementation of the BIC-plus-Laplace pipeline, which undercovers ($\text{PICP} = 0.00\text{--}0.18$); an ablation isolates the failure, showing that hard thresholding collapses accuracy while the as-implemented pipeline’s ridge-regularized fit of the ill-conditioned design matrix collapses coverage.
3. **When Sparse Selection Is Warranted.** We delineate when sparse selection is warranted: on low-dimensional data ($d = 8$) neither rule-level nor coefficient-level sparsity improves on the dense baseline, and a real high-dimensional check ($d = 81$, Supplementary Table 7) reproduces the pattern rather than overturning it—so the boundary is not a low-dimensional

87 artifact, and sparsity claims in this setting must be demonstrated rather than assumed.

88 Section 2 reviews related work. Section 3 presents the TSK model, the membership bug, and
89 the Gibbs-based Bayesian estimators. Section 4 describes the experimental setup. Section 5 reports
90 the corrected baseline, the main comparison, calibration, and the sparsity boundary. Section 6
91 discusses implications and limitations, and Section 7 concludes.

92 2. Related Work

93 2.1. Bayesian TSK Fuzzy Systems

94 Gu and colleagues introduced Bayesian TSK classifiers with conjugate Gaussian–Wishart
95 priors [6], later extended to joint structure identification [7], imbalanced classification [8], and
96 probabilistic Gaussian centralization [9]. A zero-order Bayesian TSK variant is studied in
97 [10], prediction intervals for adaptive neuro-fuzzy inference systems have been constructed
98 through interval-network training in [11], evolving fuzzy systems maintain interval predictions
99 in nonstationary environments [12], and interval-construction methods for neural predictors are
100 surveyed in [13]; none of these targets sparse high-order consequents through posterior model
101 averaging. These frameworks provide full posterior uncertainty estimates (posterior variances
102 and prediction intervals) but employ non-sparsity-inducing priors, so every rule remains active
103 regardless of relevance [14]. Our work builds directly on this line: the dense conjugate posterior
104 (Section 3.3) is the natural non-sparsity-inducing reference against which sparse Bayesian variants
105 should be judged, and we use it to validate our MCMC implementation.

106 2.2. Sparse TSK and Bayesian Variable Selection

107 On the frequentist side, Gong et al. [15] combine LASSO with TSK consequents for embedded
108 feature selection, and Zhang et al. [16] use group-LASSO regularization for nonstationary TSK
109 networks; convergence guarantees are studied in [17]. These methods yield point estimates without
110 uncertainty quantification. Exact posterior computation for Bayesian model selection is generally
111 intractable, and analytical surrogates built on information criteria or Laplace approximations have
112 well-documented limitations [18, 19]; recent work makes the error of Laplace approximations
113 quantitative and finitely computable [20]. On the Bayesian side, spike-and-slab priors [4, 5, 21]
114 have been applied to structurally sparse Bayesian neural networks via variational inference [22]
115 and to high-dimensional sensor selection in engineering prognostics [23]. Denœux’s ENNreg [24]
116 provides a belief-function alternative to probabilistic fuzzy regression with prediction intervals

but without rule selection. High-dimensional TSK systems target regimes with far more features: Bian et al. [25] introduce a multi-head TSK system for high-dimensional data, and Xue et al. [26] combine feature selection with rule extraction. Our low-dimensional benchmarks ($d = 8$) do not exercise this regime; we return to this contrast in the sparsity-boundary discussion (Section 5.4). Distribution-free alternatives such as *split* conformal prediction [27, 28] provide finite-sample coverage without posterior sampling, though split conformal requires a separate calibration set and does not decompose predictive variance into noise and model-selection components, which our BMA construction propagates explicitly. Our contribution is to implement exact (MCMC) spike-and-slab inference for TSK consequents with BMA-calibrated prediction intervals, which to our knowledge has not been developed for linear-consequent rule models, and to evaluate it against the dense baseline.

2.3. Reproducibility in Soft Computing

Reproducibility failures arising from silent implementation errors are a recognized concern in applied machine learning [29]. Within fuzzy systems, the practical consequences of an inconsistent fuzzy partition between training and inference have not, to our knowledge, been quantified. We quantify them on a reproduction of this pattern and release the corrected code and baselines publicly.

3. Methodology

3.1. TSK Fuzzy System

For input $\mathbf{x} \in \mathbb{R}^d$ and scalar target y , a TSK system with R rules is

$$y = \sum_{j=1}^R w_j(\mathbf{x}) (\beta_{j0} + \beta_{j1}x_1 + \cdots + \beta_{jd}x_d) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

where $w_j(\mathbf{x}) = \prod_{i=1}^d \mu_{A_{ji}}(x_i) / \sum_{k=1}^R \prod_{i=1}^d \mu_{A_{ki}}(x_i)$ is the normalized firing strength of rule j , and A_{ji} are Gaussian antecedent fuzzy sets

$$\mu_{A_{ji}}(x_i) = \exp\left(-\frac{(x_i - c_{ji})^2}{2s_{ji}^2}\right), \quad (2)$$

with centers c_{ji} obtained from fuzzy c -means clustering [30] and spreads s_{ji} set to the standard deviation of the training points assigned to each cluster, scaled by a factor s . Writing the weighted design matrix $\Phi \in \mathbb{R}^{n \times R(d+1)}$ and stacking consequents into $\beta \in \mathbb{R}^{R(d+1)}$, Eq. (1) becomes the linear model $\mathbf{y} = \Phi\beta + \varepsilon$.

143 3.2. The Membership-Spread Bug

144 The spreads s_{ji} in Eq. (2) define the fuzzy partition and must be identical at training and
 145 inference time. A defect we reproduce recomputes them at prediction: it calls the membership
 146 routine on query inputs while passing a placeholder label vector (e.g., all zeros), so the routine
 147 re-estimates spreads from the query data with degenerate cluster assignments. The resulting
 148 partition no longer matches the one used to fit β . In our reproduction, this single defect depresses
 149 the dense TSK baseline from $R^2 = 0.94$ to 0.48 on the Energy-Cooling target (Section 5.1).
 150 The correct procedure stores the training-time centers and spreads and reuses them verbatim at
 151 inference, as in Algorithm 1.

Algorithm 1 Corrected TSK training and inference

Require: Training data $\{\mathbf{x}_i, y_i\}_{i=1}^n$, rule count R

- 1: $(\mathbf{c}, \mathbf{u}) \leftarrow \text{FCM}(\mathbf{X}, R)$ ▷ fuzzy c -means
 - 2: $\mathbf{s} \leftarrow \{\text{per-cluster standard deviations from training points, scaled by } s\}$
 - 3: Store $(\mathbf{c}, \mathbf{s})$ ▷ fix the partition at training time
 - 4: Build Φ via Eqs. (1)–(2) using $(\mathbf{c}, \mathbf{s})$
 - 5: Fit β (least squares, conjugate Bayes, or Gibbs; Sections 3.3–3.4)
 - 6: **return** model with frozen $(\mathbf{c}, \mathbf{s}, \beta)$
-

152 3.3. Conjugate Bayesian Baseline

153 The non-sparse Bayesian TSK places a Gaussian prior $\beta \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I})$ and an inverse-gamma
 154 prior $\sigma^2 \sim \text{InvGamma}(0.01, 0.01)$ on the noise variance. The posterior is available in closed form:
 155 with $\mathbf{P} = \Phi^\top \Phi + \tau^{-2} \mathbf{I}$,

$$\beta \mid \mathcal{D}, \sigma^2 \sim \mathcal{N}(\mathbf{P}^{-1} \Phi^\top \mathbf{y}, \sigma^2 \mathbf{P}^{-1}), \quad (3)$$

156 and the predictive variance at a new design row ϕ^* is $\sigma^2 + \phi^* (\sigma^2 \mathbf{P}^{-1}) \phi^{*\top}$. This closed-form model
 157 serves both as a calibrated reference and as a correctness check for our sampler (a Gibbs sampler
 158 with all rules forced active must recover it). When the block-Gibbs sampler of Section 3.4 is
 159 run with all inclusion indicators fixed to $\gamma_j = 1$, the sampled predictive distribution recovers
 160 the conjugate closed form on held-out data: the predictive mean matches in R^2 to within 10^{-3}
 161 (measured 0.0010 at the default $R = 5$), and the predictive variance agrees to within $\sim 6\%$ (median;
 162 measured 5.6%), confirming the correctness of the β and σ^2 updates against the closed form
 163 (reproduction log in `results/raw/gibbs_verify_output.txt`). The inclusion-indicator full
 164 conditional of Eq. (5) is not exercised by this fixed-activation check; we validate it separately on a
 165 synthetic problem with known sparse ground truth, where the sampler’s inclusion probabilities

166 agree with exhaustive model averaging to within 10^{-3} (Supplementary, Section 9). We compare
 167 at the predictive level because the normal-equations matrix $\Phi^\top \Phi$ is numerically ill-conditioned
 168 (condition number $\sim 10^{18}$), which makes the raw posterior covariance numerically unstable for
 169 both the closed form and any MCMC estimate, even though the regularized estimation problem is
 170 well posed; the dense fits use ridge-regularized normal equations (penalty 10^{-4} on $\Phi^\top \Phi$), which
 171 stabilizes them.

172 3.4. Spike-and-Slab Prior and Gibbs Sampling

173 We decompose β into R blocks $\beta_j \in \mathbb{R}^{d+1}$ and place a spike-and-slab prior on each block:

$$\beta_j \mid \gamma_j, \tau^2 \sim \begin{cases} \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}_{d+1}), & \gamma_j = 1 \text{ (slab)}, \\ \delta_{\mathbf{0}}, & \gamma_j = 0 \text{ (spike)}, \end{cases} \quad (4)$$

174 with $\gamma_j \sim \text{Bernoulli}(\pi)$. Rather than approximate the posterior over γ with a BIC criterion, we
 175 sample it exactly with a block-Gibbs sampler. Here “exact” means that we target the posterior by
 176 sampling rather than replacing it with a BIC-plus-Laplace analytical surrogate; as with any MCMC
 177 procedure the estimates carry finite Monte Carlo error, which we quantify with the convergence
 178 diagnostics of Section 4. For block j , let $\mathbf{r}_{-j} = \mathbf{y} - \sum_{k \neq j} \Phi_k \beta_k$ be the residual excluding rule j , and
 179 let $\mathbf{A}_j = \Phi_j^\top \Phi_j$. The marginal likelihood of $\mathbf{r}_{-j}$ under $\gamma_j = 1$ (integrating out β_j) is Gaussian with
 180 covariance $\sigma^2 \mathbf{I} + \tau^2 \Phi_j \Phi_j^\top$, whose determinant and quadratic form are computed in $\mathcal{O}(n(d+1)^2)$
 181 via the matrix-determinant and Woodbury identities. The full conditional for γ_j is

$$P(\gamma_j = 1 \mid \cdot) = \sigma\left(\log \frac{\pi}{1-\pi} - \frac{1}{2} \log \det(\mathbf{I} + \frac{\tau^2}{\sigma^2} \mathbf{A}_j) + \frac{\tau^2}{2\sigma^4} \mathbf{z}_j^\top (\mathbf{I} + \frac{\tau^2}{\sigma^2} \mathbf{A}_j)^{-1} \mathbf{z}_j\right), \quad (5)$$

182 where $\mathbf{z}_j = \Phi_j^\top \mathbf{r}_{-j}$ and $\sigma(\cdot)$ is the logistic function. Conditional on $\gamma_j = 1$, we draw $\beta_j \sim$
 183 $\mathcal{N}((\mathbf{A}_j/\sigma^2 + \tau^{-2} \mathbf{I})^{-1} \mathbf{z}_j/\sigma^2, (\mathbf{A}_j/\sigma^2 + \tau^{-2} \mathbf{I})^{-1})$; otherwise $\beta_j = \mathbf{0}$. Finally σ^2 is drawn from its
 184 inverse-gamma full conditional. A coefficient-level variant applies the same prior independently
 185 to each scalar coefficient β_i (stochastic search variable selection, SSVS [5]), which is appropriate
 186 when feature-level rather than rule-level sparsity is of interest. For numerical stability on the
 187 ill-conditioned design matrix, the implementation adds a small diagonal jitter (10^{-10}) to the
 188 sampled precision matrices, initializes the coefficients from a ridge-regularized least-squares fit,
 189 and floors the noise variance at 10^{-4} .

190 3.5. Bayesian Model Averaging for Prediction

191 We predict by BMA rather than by conditioning on a single selected model [31]; recent
 192 reviews of BMA practice and its pitfalls are given in [32]. Given T posterior draws $\{\boldsymbol{\beta}^{(t)}, \sigma^{2(t)}\}_{t=1}^T$,
 193 the predictive mean is the mixture mean $\hat{y}^* = \frac{1}{T} \sum_t \boldsymbol{\phi}^* \boldsymbol{\beta}^{(t)}$, and the predictive variance follows
 194 from the law of total variance:

$$\text{Var}(y^* | \mathbf{x}^*, \mathcal{D}) = \underbrace{\frac{1}{T} \sum_t \sigma^{2(t)}}_{\text{aleatoric}} + \underbrace{\text{Var}_t(\boldsymbol{\phi}^* \boldsymbol{\beta}^{(t)})}_{\text{model uncertainty}}, \quad (6)$$

195 yielding the 95% interval $\hat{y}^* \pm 1.96 \sqrt{\text{Var}(y^* | \mathbf{x}^*, \mathcal{D})}$. Over posterior draws the second term spans
 196 both between-model (rule-configuration) variation and within-model parameter variation; it is
 197 precisely the model-selection uncertainty that a hard-thresholded approximation discards, and
 198 retaining it is what restores calibration.

199 4. Experimental Setup

200 We evaluate on two UCI regression benchmarks (three targets): the Energy Efficiency dataset
 201 [2] (768 simulated building configurations, 8 geometric features, with heating and cooling loads
 202 as two targets) and the Concrete Compressive Strength dataset [33] (1030 laboratory samples, 8
 203 mix-design features). Features are standardized to zero mean and unit variance; all TSK methods
 204 use $R = 5$ rules from fuzzy c -means ($m = 2$) with Gaussian membership spreads equal to the
 205 per-cluster standard deviation of training points, scaled by $s = 1.5$. A rule-count sweep ($R = 3$ –10,
 206 Supplementary) shows dense accuracy is essentially flat from $R = 4$ upward on the Energy targets
 207 and improves only marginally beyond $R \approx 7$ on Concrete, supporting the default $R = 5$.

208 Four TSK methods are compared in the main experiments: **TSK-LS** (dense least squares),
 209 **Bayesian-TSK** (dense conjugate posterior, Section 3.3), **TSK-SpikeSlab-BIC** (the BIC-plus-
 210 Laplace analytical approximation with hard thresholding; labeled SpikeSlab-Fast in the re-
 211 leased code), and **TSK-SpikeSlab-Gibbs** (rule-level Gibbs plus BMA, Sections 3.4–3.5; labeled
 212 SpikeSlab-Gibbs); a coefficient-level variant, **TSK-SSVS** (labeled SSVS-Gibbs), is used in
 213 the sparsity-boundary analysis (Section 5.4). Random forest, support vector regression, and
 214 Gaussian-process regression serve as non-interpretable machine-learning references; the Gaussian
 215 process provides closed-form predictive intervals and is therefore the key probabilistic baseline for
 216 the calibration comparison. Reference implementations use the scikit-learn configurations shipped

217 in the released code: a random forest with 300 trees and maximum depth 10, support vector
 218 regression with an RBF kernel ($C = 1$, $\gamma = \text{scale}$), and a Gaussian process with an RBF-plus-
 219 white-noise kernel, hyperparameters optimized by marginal likelihood (`normalize_y=True`).
 220 Fuzzy-specific interval-construction methods such as ENNreg [24] and interval-ANFIS [11]
 221 build prediction intervals through evidential combination or interval-network training rather than
 222 posterior model selection; they are therefore not direct baselines for the model-selection question
 223 addressed here, and the Gaussian process—whose closed-form intervals come from a standard
 224 library implementation—serves as the probabilistic reference. The rule-level Gibbs sampler uses
 225 1,000 burn-in and 2,000 retained draws; the coefficient-level SSVS uses 800 and 800. For the
 226 rule-level sampler, convergence is verified by running three chains with independent seeds on
 227 representative splits: the Gelman–Rubin statistic satisfies $\hat{R} \leq 1.001$ for the consequents and the
 228 noise variance, and the effective sample size exceeds 1,700 for the noise variance (Supplementary
 229 Table 6). The inclusion-indicator chains saturate near full activation on these benchmarks (Sec-
 230 tion 5.4), so their effective sample size equals the chain length by construction rather than from
 231 active exploration; formal convergence verification is claimed only for the rule-level sampler,
 232 not for the coefficient-level SSVS runs. All results are averaged over 30 independent random
 233 80/20 train–test splits with a fixed seed (SEED=42); reported intervals are 95% Gaussian intervals.
 234 All spike-and-slab priors use inclusion probability $\pi = 0.5$; the rule-level Gibbs and conjugate
 235 baselines use slab variance $\tau^2 = 10^3$, and the coefficient-level SSVS uses $\tau^2 = 1.0$. The $\pi = 0.5$
 236 prior is dominated by the data on these benchmarks (posterior inclusion probabilities saturate near
 237 one), and the rule-level slab $\tau^2 = 10^3$ is deliberately diffuse relative to the standardized-feature
 238 scale; the sensitivity of the coefficient-level variant to τ^2 is characterized in Fig. 6a. Metrics are
 239 root-mean-square error (RMSE), R^2 , prediction-interval coverage probability (PICP), and mean
 240 prediction-interval width (MPIW).

241 5. Results

242 5.1. A Corrected Dense Baseline

243 Fig. 1 contrasts the R^2 values reproduced under the buggy membership implementation with
 244 those obtained after correcting it. The improvement is consistent: TSK-LS rises from 0.48 to 0.94
 245 (Energy-Cooling), 0.78 to 0.97 (Energy-Heating), and -2.86 to 0.74 (Concrete); the conjugate
 246 Bayesian-TSK rises from 0.78 to 0.94, 0.84 to 0.97, and -0.56 to 0.74 . This confirms that the

bug, not the model class, was responsible for the apparently mediocre baselines, and it establishes the corrected numbers as the reference for the comparisons below.

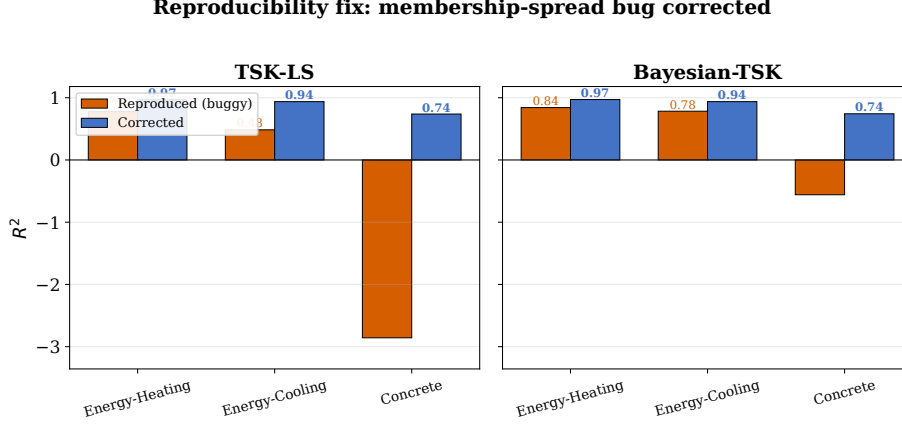


Figure 1: Corrected dense baselines. Reproduced R^2 under the buggy membership implementation (orange) versus corrected R^2 (blue, training-time membership spreads reused at inference) for the dense TSK-LS and conjugate Bayesian-TSK baselines. Values are means over 30 splits on the Energy (Heating and Cooling) and Concrete targets; correcting the membership-spread bug raises the dense TSK-LS R^2 from 0.48 to 0.94 on Energy-Cooling.

5.2. Main Comparison

Table 1 and Fig. 2 report the primary results (full mean-absolute-error results in Supplementary Table 1). After correcting the bug, the dense baselines are strong: TSK-LS achieves $R^2 = 0.94$ – 0.97 on the Energy targets and 0.74 on Concrete, and the conjugate Bayesian-TSK differs from it by at most 0.01 in R^2 . The proposed Gibbs sampler recovers this accuracy: TSK-SpikeSlab-Gibbs attains $R^2 = 0.97/0.94/0.74$, matching the dense baseline to within 0.01 in R^2 on every target, which is the expected behavior of a correct sampler and a direct validation of our MCMC implementation against the closed form. A paired Wilcoxon signed-rank test over the 30 splits [34] finds no significant difference between TSK-SpikeSlab-Gibbs and TSK-LS on the Energy targets (Heating $p = 0.11$, Cooling $p = 0.07$) or between Bayesian-TSK and TSK-LS (Heating $p = 0.32$, Cooling $p = 0.87$); the test does not establish equivalence, and the split-level results are not independent because the splits reuse the same underlying rows. On Concrete the difference is nominally significant ($p < 0.01$) but tiny in magnitude, with the samplers slightly above the dense baseline (mean $\Delta R^2 = 0.006$ for Gibbs, 0.004 for Bayesian-TSK).

By contrast, the BIC-based analytical approximation TSK-SpikeSlab-BIC collapses to negative R^2 (-1.8 to -6.0), significantly worse than every other method on all three targets (all $p < 0.001$);

it predicts worse than a constant mean because hard thresholding discards rules that the correctly specified model needs.

The Gaussian-process reference [35] attains the accuracy of the non-interpretable Random Forest (GP $R^2 = 0.998/0.98/0.89$) but its closed-form 95% intervals slightly under-cover (PICP = 0.93–0.94), whereas the exact Bayesian TSK methods reach near-nominal coverage (PICP = 0.93–0.94), whereas the exact Bayesian TSK methods reach near-nominal coverage (PICP = 0.93–0.95) at the cost of wider intervals; on the Heating and Concrete targets the TSK intervals sit closer to the nominal level than the GP’s, while on Cooling the two are essentially tied (TSK 0.93 vs GP 0.94).

Table 1: Main comparison (mean over 30 splits; $\pm$ std shown for RMSE; R^2 and PICP are means). PICP is the 95% interval coverage. Differences among TSK-LS, Bayesian-TSK, and TSK-SpikeSlab-Gibbs are at most 0.01 in R^2 ; paired Wilcoxon tests over the 30 splits are reported in the text.

Method	Energy-Heating			Energy-Cooling			Concrete		
	RMSE	R^2	PICP	RMSE	R^2	PICP	RMSE	R^2	PICP
TSK-LS	1.69 $\pm$ 0.19	0.97	–	2.36 $\pm$ 0.17	0.94	–	8.52 $\pm$ 0.55	0.74	–
Bayesian-TSK	1.69 $\pm$ 0.19	0.97	0.95	2.36 $\pm$ 0.17	0.94	0.93	8.46 $\pm$ 0.43	0.74	0.95
TSK-SpikeSlab-BIC	16.4 $\pm$ 3.4	–1.8	0.00	18.8 $\pm$ 3.0	–3.1	0.00	44.0 $\pm$ 4.0	–6.0	0.18
TSK-SpikeSlab-Gibbs	1.70 $\pm$ 0.19	0.97	0.95	2.37 $\pm$ 0.17	0.94	0.93	8.43 $\pm$ 0.43	0.74	0.95
Random Forest	0.49 $\pm$ 0.06	0.998	–	1.67 $\pm$ 0.13	0.97	–	5.17 $\pm$ 0.40	0.90	–
SVR	2.73 $\pm$ 0.21	0.93	–	3.09 $\pm$ 0.22	0.89	–	9.97 $\pm$ 0.43	0.64	–
Gaussian Process	0.47 $\pm$ 0.05	0.998	0.93	1.35 $\pm$ 0.11	0.98	0.94	5.53 $\pm$ 0.46	0.89	0.94

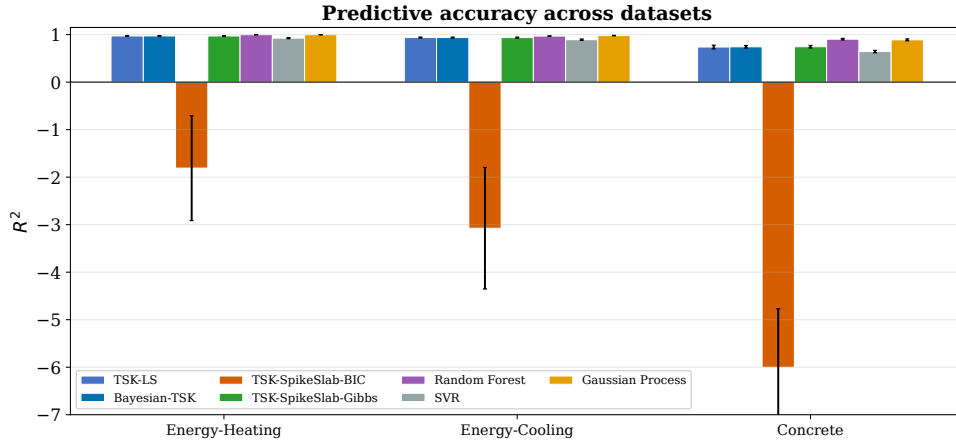


Figure 2: Predictive accuracy (R^2) across datasets. The Gibbs/BMA sampler (TSK-SpikeSlab-Gibbs) recovers the dense-baseline accuracy on all three targets (Energy-Heating, Energy-Cooling, Concrete; $R^2 = 0.97/0.94/0.74$), while the BIC-based analytical approximation (TSK-SpikeSlab-BIC) collapses to negative R^2 .

273 5.3. Calibration

274 Fig. 3 reports interval calibration, assessed with the coverage and width diagnostics standard
 275 in the probabilistic-forecast literature [36]. Pooled over the 30 splits, both the conjugate Bayesian-
 276 TSK and the Gibbs sampler attain mean PICP of 0.95 on the Energy-Heating and Concrete
 277 targets and 0.935 on Energy-Cooling (displayed as 0.93 in Table 1), with 95% binomial (Wilson)
 278 intervals $[0.945, 0.958]$, $[0.944, 0.955]$, and $[0.927, 0.941]$, respectively (the Gibbs values agree
 279 with the conjugate values to within 0.004). Coverage is therefore within one binomial standard
 280 error of the nominal 95% on Heating and Concrete, and about 1.5 percentage points below it on
 281 Cooling—a deviation that lies within one per-split standard deviation (~ 0.027 on Cooling) yet is
 282 statistically detectable once the splits are pooled—so we describe the intervals as near-nominal
 283 rather than perfectly calibrated. Widths are compact ($\text{MPIW} \approx 6.7\text{--}32.9$; Supplementary Table
 284 8). The analytical TSK-SpikeSlab-BIC baseline undercovers severely as implemented: $\text{PICP} = 0.00$ on both Energy targets and 0.18 on Concrete, meaning essentially all test points on the
 285 Energy targets, and the large majority on Concrete, fall outside the nominal 95% intervals. The
 286 ablation (Supplementary, Section 5) isolates the mechanism: hard thresholding alone collapses
 287 accuracy ($R^2 = 0.60$) while retaining near-nominal coverage with inflated width ($\text{MPIW} 26.1$),
 288 so the coverage collapse arises from the as-implemented pipeline’s ridge-regularized fit of the
 289 ill-conditioned design matrix rather than from the Laplace covariance per se; the Gibbs/BMA
 290 prediction of Eq. (6) restores accuracy and efficiency, propagating model uncertainty through
 291 the mixture variance. The value of averaging over configurations is most visible where the
 292 closed-form conjugate baseline degrades: on the high-dimensional check (Superconductivity,
 293 $d = 81$, Section 5.4, Supplementary Table 7), the Gibbs/BMA estimator maintains near-nominal
 294 calibration ($\text{PICP} 0.940$) where the dense conjugate baseline slightly undercovers ($\text{PICP} 0.921$).
 295 Fig. 4 confirms this calibration across a range of nominal levels, and Fig. 5 shows the 95%
 296 intervals of the two Bayesian methods against the held-out truth on a representative split.
 297

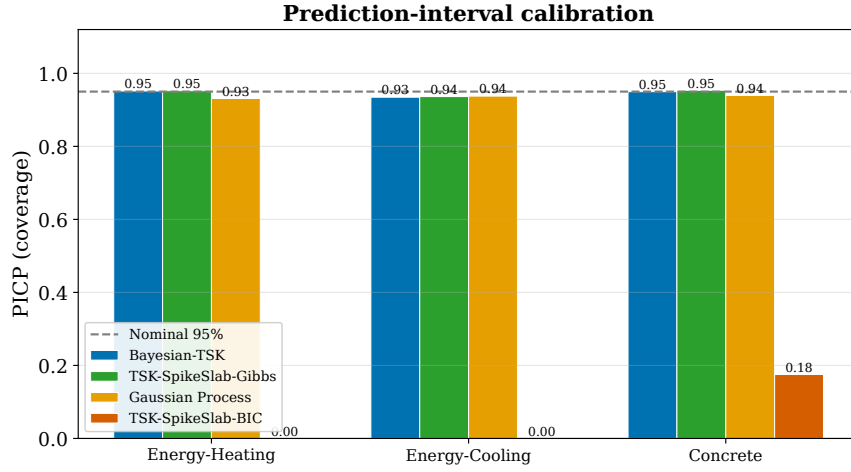


Figure 3: Prediction-interval coverage (PICP) with the nominal 95% reference (dashed). Gibbs/BMA and the conjugate Bayesian model attain near-nominal coverage; the BIC-plus-Laplace approximation undercovers severely as implemented.

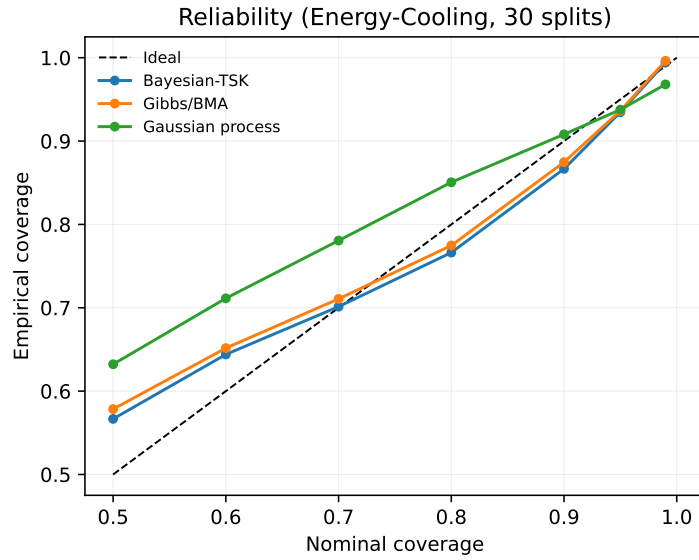


Figure 4: Empirical coverage versus nominal coverage for the conjugate Bayesian-TSK, the Gibbs/BMA sampler, and the Gaussian-process reference on Energy-Cooling (30 splits). Points on the diagonal would indicate perfectly calibrated intervals. The Bayesian methods over-cover at the lowest nominal levels (empirical coverage 0.57 at nominal 0.5), under-cover slightly in the range 0.8–0.95 (empirical 0.94 at nominal 0.95), and return to the diagonal at nominal 0.99, while the Gaussian process falls below the diagonal at nominal 0.99.

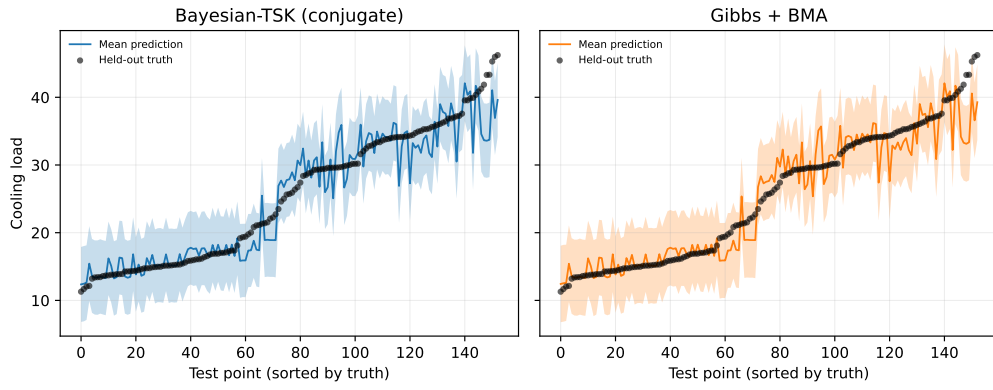


Figure 5: 95% prediction intervals (shaded) and mean prediction (line) versus held-out truth (points) on one representative Energy-Cooling split, for the conjugate Bayesian-TSK (left) and Gibbs+BMA (right). The intervals track the sorted truth closely and cover the large majority of test points at compact widths.

298 5.4. The Sparsity Boundary

299 Having established that Gibbs/BMA is correct and calibrated, we characterize when the
300 *sparsity* component of the spike-and-slab prior helps. Two results stand out (Fig. 6).

301 First, rule-level sparsity rarely activates: on all three targets the posterior inclusion probabilities
302 of the $R = 5$ rules are close to 1.0 (with a single rule occasionally pruned on the Energy targets),
303 because with a correct partition nearly every rule contributes. This is expected in the low-
304 dimensional regime ($d = 8$), where the dense estimation problem is well posed; the rule-count
305 sweep (Supplementary) confirms that even at $R = 10$ the inclusion probabilities remain near 1.0,
306 so low dimensionality, not the rule count, keeps rule-level sparsity inactive.

307 Second, coefficient-level sparsity (SSVS) is a trade-off, not an accuracy gain. Varying the slab
308 variance τ^2 over three orders of magnitude (Fig. 6a) shows that a tight slab ($\tau^2 = 0.1$) over-shrinks
309 and collapses accuracy to $R^2 = -5.4$, whereas a diffuse slab ($\tau^2 = 100$) merely recovers the
310 dense baseline ($R^2 = 0.92$ versus 0.94) while pruning nothing. No τ^2 setting improves on the
311 dense baseline. In the irrelevant-feature regime (Fig. 6b), appending up to 30 standard-normal
312 noise features leaves the dense baseline near its original level (ending at $R^2 = 0.88$, with a small
313 non-monotone rise at four features that is within split-level noise), and the sparse variants do not
314 improve upon it. Correct fuzzy c -means already provides robustness to irrelevant features, so
315 sparsity does not improve on the dense baseline in this low-dimensional regime.

Sparsity boundary under correct fuzzy c-means

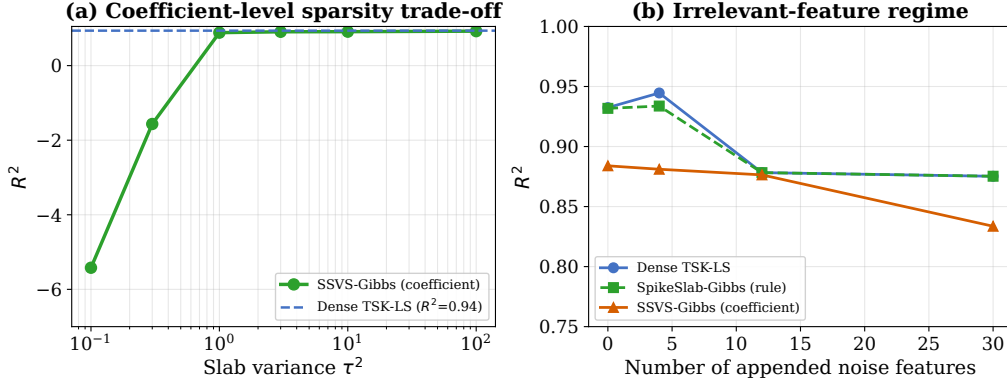


Figure 6: The sparsity boundary on Energy-Cooling ($R = 5$). (a) Coefficient-level SSVS R^2 versus slab variance τ^2 (10 splits): a tight slab over-shrinks and a diffuse slab only recovers the dense baseline (dashed; 30-split TSK-LS reference, $R^2 = 0.94$), never improving on it. (b) Irrelevant-feature regime (10 splits; dense baseline $R^2 = 0.93$ at zero noise features): the dense TSK-LS declines only modestly, from $R^2 = 0.93$ to 0.88, as noise features are appended (a small non-monotone rise at four features is within split-level noise), and neither rule-level nor coefficient-level sparsity improves on it (details in Supplementary Tables 2–3).

6. Discussion

6.1. Why the Analytical Approximation Fails

The failure of the BIC-plus-Laplace approximation has a specific origin. BIC approximates the Bayes factor [37] by replacing the integral over the slab with a point mass at the maximum-likelihood estimate; this is only reliable when the posterior over γ_j is concentrated at 0 or 1. On the Energy benchmark the inclusion probabilities are intermediate, so the hard threshold at 0.5 discards rules that carry real (if partial) signal. The Laplace predictive covariance then conditions on a single pruned model; with an ordinary-least-squares fit it alone yields wide intervals with near-nominal coverage (MPIW 26.1, PICP 0.93), so the coverage collapse instead arises in the as-implemented pipeline, which fits the selected model with ridge regularization on the ill-conditioned design matrix. Gibbs sampling with BMA avoids both steps: it averages over all rule configurations rather than selecting one. An ablation isolates which step is decisive (Supplementary, Section 5): replacing hard thresholding with BMA over all 2^R rule subsets, using the same BIC-based model weights, restores dense-level accuracy and near-nominal calibration ($R^2 = 0.95$, PICP 0.92 on Energy-Cooling), so averaging rather than the BIC approximation is the decisive change for accuracy. Feeding the Gibbs sampler’s inclusion probabilities through

332 hard thresholding and the Laplace covariance performs equally well ($R^2 = 0.94$, PICP 0.94),
 333 confirming that the Gibbs inclusion probabilities are reliable; the as-implemented BIC baseline
 334 collapses ($R^2 = -2.5$, PICP 0.00; 10-split ablation, Supplementary Section 5, cf. -3.1 at 30 splits
 335 in Table 1), with the coverage collapse traceable to its ridge-regularized fit of the ill-conditioned
 336 design matrix.

337 6.2. Implications for Sparse TSK Research

338 Our boundary characterization carries a constructive message. Sparsity-inducing priors are
 339 worth their cost only when the model is over-parameterized (high-dimensional inputs with truly
 340 irrelevant features) or when the data are scarce relative to the number of effective parameters;
 341 our real high-dimensional check ($d = 81$, Supplementary Table 7) does not realize this regime,
 342 since its features are largely relevant and $n = 3,000$ is not scarce relative to the effective parameter
 343 count. In the low-dimensional building-energy regime, the dense TSK is already well posed and
 344 no sparsity is warranted. The rule-count sweep (Supplementary) sharpens this: even at $R = 10$,
 345 where the model has $R(d+1) = 90$ consequent coefficients, rule-level sparsity does not activate, so
 346 rule count alone, without high dimensionality, does not make sparse selection advantageous on
 347 these benchmarks. This is not a failure of the spike-and-slab framework; it is a statement about the
 348 data, and it pinpoints where the framework’s cost is justified. Alternative shrinkage priors, such
 349 as the continuous spike-and-slab LASSO [38] and the horseshoe [39], offer smooth regularizers
 350 that may behave differently under the collinearity of the TSK weighted design matrix; extending
 351 the exact inference developed here to these priors is natural future work. We therefore present
 352 the Gibbs/BMA estimator as a *correct and calibrated* Bayesian TSK, and report sparsity as a
 353 boundary condition rather than a primary contribution.

354 6.3. Limitations

355 Three limitations scope this work. First, the benchmarks are low-dimensional ($d = 8$), although
 356 an evaluation on a real high-dimensional benchmark (Superconductivity, $d = 81$, subsample of
 357 $n = 3,000$ rows, Supplementary Table 7) reproduces the same pattern—rule-level sparsity matches
 358 but does not improve on the dense baseline, and coefficient-level sparsity over-shrinks—so the
 359 boundary is not a low-dimensional artifact; whether sparsity is beneficial under different antecedent
 360 structures or much larger samples remains open. Second, we fix the antecedent structure and
 361 apply sparsity only to consequents; joint antecedent–consequent selection [7] remains future work.

362 Third, our evaluation uses simulated (Ecotect) building-energy data; validation against measured,
363 submetered energy data would strengthen the application case.

364 7. Conclusion

365 We introduced exact Bayesian inference for spike-and-slab priors over TSK consequents:
366 a block-Gibbs sampler combined with Bayesian model averaging that recovers near-nominal
367 95% prediction intervals ($\text{PICP} = 0.93\text{--}0.95$) where our implementation of the BIC-plus-Laplace
368 analytical pipeline undercovers ($\text{PICP} = 0.00\text{--}0.18$). In practical terms, a modeler who needs a
369 95% interval from a rule-based model should average over rule configurations rather than commit
370 to a single selected model: the averaged intervals track the nominal level closely, whereas the
371 thresholded BIC baseline undercovers severely. We validated the method against a corrected
372 dense baseline, obtained by fixing a membership-function bug that had depressed TSK baselines
373 ($R^2 = 0.48 \rightarrow 0.94$ on the Energy-Cooling target). We also characterized when sparsity helps,
374 showing that on low-dimensional data neither rule-level nor coefficient-level sparsity improves on
375 the dense baseline, delineating the regime in which sparse selection is warranted. The approach
376 applies to any rule-based regression with linear consequents, and the corrected code and baselines
377 are released to support reproducible, uncertainty-aware inference.

378 CRediT authorship contribution statement

379 **Zhiren Xiao:** Conceptualization, Methodology, Software, Validation, Formal analysis, Inves-
380 tigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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383 Data Availability

384 Datasets are publicly available from the UCI Machine Learning Repository: Energy Efficiency
385 (dataset 242) [2] and Concrete Compressive Strength (dataset 165) [33]. Code is available at
386 <https://github.com/Jackxiaozhiren/tsk-spikeslab> and archived with the persistent
387 DOI [10.5281/zenodo.21929319](https://doi.org/10.5281/zenodo.21929319), including the corrected TSK core, experiment drivers, a pinned
388 dependency manifest, and the fixed seed stated in Section 4.

389 Declaration of Competing Interests

390 The author declares no competing interests.

391 Declaration of Generative AI and AI-assisted technologies in the writing process

392 During the preparation of this work the author used Claude (Anthropic) for code development
393 assistance and language editing. All scientific content, methodology design, experimental results,
394 and interpretation were performed and verified by the author.

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